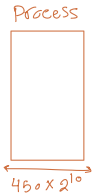


## Os midterm fall 2022

### Q.1 (50%): Memory Management

- A. Consider a system of 16 GB ( $2^{34}$  B) of memory with a frame size of 8 KB ( $2^{13}$  B) and a process P with a logical address space of size 450 KB. Assume that page N of this process is mapped to frame N+2000. Answer the following questions showing your work.



1. What is the physical address of the logical address 345678?

$LA: 345678$   
 $VLA: (\frac{345678}{2^{13}} = 42, 345678 - (42 \times 2^{13}) = 1614)$   
 $VPA: (2000 + 42 = 2042, 1614)$   
 $PA: 2042 \times 2^{13} + 1614 = 2042 \times 2^{13} + 1614 = 16,729,678$

$Memory = 2^{34}$   
 $Frame\ size = 2^{13}$   
 $Process\ size = 450 \times 2^{10}$

2. What is the amount of internal fragmentation, if any?

$57 \times 2^{13} - 450 \times 2^{10} = 6144B$

3. How many pages are needed to represent this process?

$\frac{450}{8KB} = \frac{450 \times 2^{10}}{2^{13}} = 56.2 \Rightarrow 57$

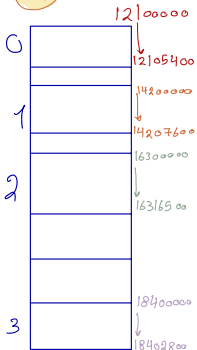
4. How many bits are needed to represent the offset?

$8KB = 2^3 \times 2^{10} = 2^{13} \rightarrow 13\ bits$

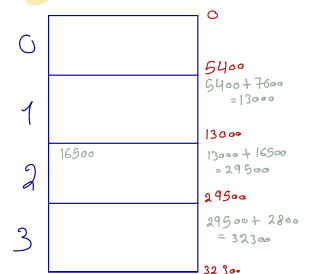
- B. Consider the following segmentation table for the process P with zero internal fragmentation. Answer the following questions showing your work.

| Segment Number | Base Address | Limit |
|----------------|--------------|-------|
| 0              | 12100000     | 5400  |
| 1              | 14200000     | 7600  |
| 2              | 16300000     | 16500 |
| 3              | 18400000     | 2800  |

2 Draw memory size



1 Draw Process



1. What is the physical address of the logical address 16500?

$LA: 16500$   
 $VLA: (Seg\#2, 16500 - 13000 = 3500)$   
 $PA: 16300000 + 3500 = 16303500$

2. What is the virtual logical address of the physical address 14204567?

$PA: 14204567 \Rightarrow (Seg\#1, offset?)$   
 $(Seg\#1, 14204567 - 14200000 = 4567)$

3. What is the total size of the process address space?

$5400 + 7600 + 16500 + 2800 = 32300$

C. Consider a memory of size 32KB) This memory is allocated in increments of 2KB and the memory is currently represented as follows:

|               |                   |                |                   |               |
|---------------|-------------------|----------------|-------------------|---------------|
| Free<br>(6KB) | Occupied<br>(8KB) | Free<br>(10KB) | Occupied<br>(4KB) | Free<br>(4KB) |
|---------------|-------------------|----------------|-------------------|---------------|

Use the various contiguous allocation schemes (First-fit, Best-fit, and Worst-fit) to show the memory representation after satisfying the allocation requests for processes requesting sizes as given below.

|                        |     |     |     |     |
|------------------------|-----|-----|-----|-----|
| Process                | A   | B   | C   | D   |
| Requested size (in KB) | 3.5 | 2.3 | 5.2 | 1.6 |

4

4

6

2

|                                                                                                                                    |  |
|------------------------------------------------------------------------------------------------------------------------------------|--|
| First-fit                                                                                                                          |  |
| Best-fit<br><i>internal Fragmentation:</i><br>$A: 4 - 3.5 = 0.5$<br>$B: 4 - 2.3 = 1.7$<br>$C: 6 - 5.2 = 0.8$<br>$D: 2 - 1.6 = 0.4$ |  |
| Worst-fit                                                                                                                          |  |
| The amount of internal fragmentation (in KB) for Best-fit algorithm is <u><math>0.5 + 1.7 + 0.8 + 0.4 = 3.4</math> KB</u> .        |  |
| The amount of external fragmentation (in KB) for Worst-fit algorithm is <u><math>2\text{KB} + 2\text{KB} = 4\text{KB}</math></u> . |  |

- D. Consider a system in which a process is allocated three frames in memory. Given the page requests of this process as "7, 5, 3, 1, 5, 2, 4, 6, 4, 2, 5, 7, 1, 5, 3", trace the utilization of the three frames by this process and find the number of page faults when using the Least Recently Used (LRU) replacement algorithm.

| Frame | 7 | 5 | 3 | 1 | 5 | 2 | 4 | 6 | 4 | 2 | 5 | 7 | 1 | 5 | 3 |
|-------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0     | 7 | 7 | 7 | 1 |   | 1 | 4 | 4 |   |   | 4 | 7 | 7 |   | 3 |
| 1     |   | 5 | 5 | 5 |   | 5 | 5 | 6 |   |   | 5 | 5 | 5 |   | 5 |
| 2     |   |   | 3 | 3 |   | 2 | 2 | 2 |   |   | 2 | 2 | 1 |   | 1 |

Number of page Faults = 11

$$\text{Effective Access Time} = ((1-P) * \text{Memory Access Time}) + (P * \text{Avg Fault service Time})$$

total = 15

fault = 11

## Q.2 (30%): Multiple Choice

1. What is the output of the following program?

```
#include<stdio.h> #include<stdlib.h>
#include <unistd.h>
```

→ int main() {

```
    int pid_t child;
    child Parent
```

```
    int status;
```

```
    Child=fork();
```

```
    switch=(child) {
```

```
    case -1:
```

```
        perror("fork");
        exit(1);
```

```
    case 0:
```

```
        child exit (2); → status = 2
        break;
```

```
    default:
```

```
        while (waitpid (child, &status, WNOHANG) == 0)
            sleep(1);
        printf("%d\n", WEXITSTATUS (status));
    }
```

```
    return 0;
```

a) 0

b) 1

c) 2

d) None of the above

fork()



while child is alive

WNOHANG → == 0  
means check if child alive or not

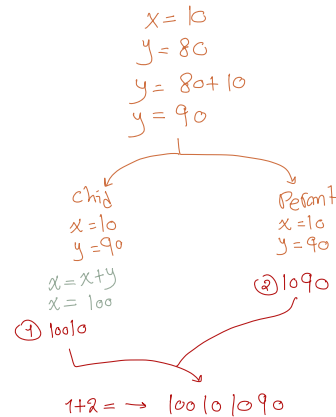
exit code for the child  
= 2

2. What is the output of the following program?

```
#include<stdio.h>
#include <unistd.h>
int main() {
    int x = 10, y = 80;
    y = y + x;
    pid_t c = fork();
    if (c == 0) {
        x += y;
        printf("%d%d", x, x - y);
    }
    else {
        wait (NULL);
        printf("%d%d", x, y);
    }
    return 0;
}
```

*Handwritten notes:*

- child:**  $x = x + y$  (points to  $x = 100$  in the code)
- Parent:**  $x = 10, y = 90$  (points to  $x = 10, y = 90$  in the code)



- a) 100109090
- b) 100101090**
- c) 109010010
- d) 1001010090

3. Assume didn't explain it ☺

4. How many times "CMPS405" will be printed in this program?

```
#include<stdio.h>
#include<stdlib.h>
#include <unistd.h>
int main() {
    if (fork() == 0)
        if (execl("/bin/ls", "ls", NULL) == -1) {
            perror("execl");
            exit(1);
        }
    printf("CMPS405\n");
    return 0;
}
```

*Handwritten notes:*

- child:** (points to the if (fork() == 0) block)

- a) 0**
- b) 1
- c) 2
- d) None of the above

5. In the Additional-Reference-Bits Algorithm used by LRU, which of the following shift register values represents a page that would be the best choice for replacement?

- a) 00110011
- b) 01010101
- c) 10101010
- d) 11001100

Additional Reference bit start with zero

Second chance

Enhanced Second Chance

6. Which of the following is an advantage of the Layered Structure of OS Design?

Disadvantage

- a) Defining appropriate layers data and operations is an easy task
- b) Direct user access to low-level facilities is allowed
- c) Very efficient in communicating requests between the layers
- d) Simplification of debugging and system verification ✓

7. Which type of Operating System architecture allows all processors to communicate with each other by a shared memory?

- a) Symmetric multiprocessing ✓
- b) Asymmetric multiprocessing
- c) Real Time
- d) Distributed

✗ Which of the following describes a Zombie process?

- a) A process that its parent exited while it is still running
- b) A process that makes exit when its parent is not executing wait ✓
- c) A process that is adopted by the system init process
- d) A process that replaces its memory space with a new program

✗ Which of the following is not true about the pipe operation?

- a) The file descriptor 0 is used for reading while the 1 is used for writing
- b) If a write is made on a pipe that is full the data will be dropped
- c) If a pipe is empty the reader process suspended until more output is available
- d) It requires two pipes to implement the command line `ls | sort | wc -c`

### Extra questions to solve

**Q3. [12 points]** Consider a system in which a process is allocated three frames in memory. Given the page requests of this process as “1, 4, 5, 3, 4, 5, 6, 1, 7, 5, 6, 1, 3, 2, 6, 5, 2, 1, 5, 2”, trace the utilization of the three frames by this process when using the Optimal, FIFO, LRU replacement algorithms. To compare the efficiency of using each algorithm, find the total number of page faults and the effective access time. Assume that the memory access time is 100 nanoseconds and the average page-fault service time is 4 microseconds.

[Grading: 4 points for each answer (2 for the frames, 1 for number of faults and 1 for EAT)]

#### Optimal Replacement

|   | 1 | 4 | 5 | 3 | <del>4</del> | <del>5</del> | 6 | 1 | 7 | <del>5</del> | <del>6</del> | 1 | <u>3</u> | 2 | <del>6</del> | <del>5</del> | <del>2</del> | 1 | <del>5</del> | <del>2</del> |
|---|---|---|---|---|--------------|--------------|---|---|---|--------------|--------------|---|----------|---|--------------|--------------|--------------|---|--------------|--------------|
| 0 | 1 | 1 | 1 | 3 |              |              | 3 | 1 | 7 |              |              | 7 | 3        | 2 |              |              |              | 2 |              |              |
| 1 |   | 4 | 4 | 4 |              |              | 6 | 6 | 6 |              |              | 6 | 6        | 6 |              |              |              | 1 |              |              |
| 2 |   |   | 5 | 5 |              |              | 5 | 5 | 5 |              |              | 5 | 5        | 5 |              |              |              | 5 |              |              |

Number of page faults = 11

Effective access time =  $((1-P) * MAT) + (P * AST)$

$$P = \frac{11}{20} = 0.55$$

$$((1 - 0.55) * 100 \times 10^{-9}) + (0.55 * 4 \times 10^{-6}) = 2.245 \times 10^{-6}$$

#### FIFO Replacement

|   | 1 | 4 | 5 | 3 | <del>4</del> | <del>5</del> | 6 | 1 | 7 | 5 | 6 | 1 | 3 | 2 | 6 | 5 | <del>2</del> | 1 | <del>5</del> | 2 |
|---|---|---|---|---|--------------|--------------|---|---|---|---|---|---|---|---|---|---|--------------|---|--------------|---|
| 0 | 1 | 1 | 1 | 3 |              |              | 3 | 3 | 7 | 7 | 7 | 1 | 1 | 1 | 6 | 6 |              | 6 |              | 2 |
| 1 |   | 4 | 4 | 4 |              |              | 6 | 6 | 6 | 5 | 5 | 5 | 3 | 3 | 3 | 5 |              | 5 |              | 5 |
| 2 |   |   | 5 | 5 |              |              | 5 | 1 | 1 | 1 | 6 | 6 | 6 | 2 | 2 | 2 |              | 1 |              | 1 |

Number of page faults = 16

Effective access time =  $((1 - 0.8) * 100 \times 10^{-9}) + (0.8 * 4 \times 10^{-6})$

$$\frac{16}{20} = 0.8$$

#### LRU Replacement

|   | 1 | 4 | 5 | 3 | <del>4</del> | <del>5</del> | 6 | 1 | 7 | 5 | 6 | 1 | 3 | 2 | 6 | 5 | <del>2</del> | 1 | <del>5</del> | <del>2</del> |
|---|---|---|---|---|--------------|--------------|---|---|---|---|---|---|---|---|---|---|--------------|---|--------------|--------------|
| 0 | 1 | 1 | 1 | 3 |              |              | 6 | 6 | 6 | 5 | 5 | 5 | 3 | 3 | 3 | 5 |              | 5 |              |              |
| 1 |   | 4 | 4 | 4 |              |              | 4 | 1 | 1 | 1 | 6 | 6 | 6 | 2 | 2 | 2 |              | 2 |              |              |
| 2 |   |   | 5 | 5 |              |              | 5 | 5 | 7 | 7 | 7 | 1 | 1 | 1 | 6 | 6 |              | 1 |              |              |

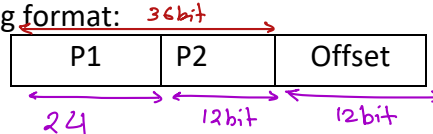
Number of page faults = 15

Effective access time =  $((1 - 0.75) * 100 \times 10^{-9}) + (0.75 * 4 \times 10^{-6})$

$$P = \frac{15}{20} = 0.75$$

Q2. [8 points] Consider a 48-bit virtual address space. The system uses a page size of 4KB and two-level hierarchical paging. Therefore, the page table is divided into 4KB pages and the logical address has the following format:

$$\frac{2^{48}}{2^{12}} = 2^{36} = 36 \text{ bit}$$



$$\frac{2^{36}}{2^{12}} = 2^{24} = 24 \text{ bits}$$

$$2^2 \times 2^{10} = 2^{12} = 12 \text{ bit offset}$$

Below are the relevant tables. The values are provided in hexadecimal.

| Outer Page Table |                  |
|------------------|------------------|
| 0                | Page A of PT     |
| 1                | Page b of PT     |
| x                | Rest are invalid |
|                  |                  |

| Page A of the Page Table |                  |
|--------------------------|------------------|
| 0                        | FEE              |
| 1                        | 3BC              |
| 2                        | 124              |
| 3                        | C3D              |
| x                        | Rest are invalid |

| Page B of the Page Table |                  |
|--------------------------|------------------|
| 0                        | 330              |
| 1                        | 44B              |
| 2                        | 355              |
| x                        | Rest are invalid |

[Grading: 1 point for each answer]

- a) [3 points] What is the size (number of bits) of each of the following logical address fields:

Size of P1 =  $36 - 12 = 24$  bits

Size of P2 =  $24 - 12 = 12$  bits

Size of Offset =  $4 \text{ KB} = 2^2 \times 2^{10} = 2^{12}$  bits

- b) [5 points] what is the physical/logical address corresponding to each of the following shown logical/physical addresses. Answer "Bad logical address" if the logical address is invalid.

| Logical Address                     | Physical Address |
|-------------------------------------|------------------|
| <u>00000000</u> <u>374F</u><br>P1 P | C3D 74F          |
| <u>00000200</u> <u>15A2</u><br>P1 P | Invalid location |
| 00001002FEE                         | E55FEE           |
| <u>00000100</u> <u>1321</u><br>P1 P | 44B 321          |
| 00000000330                         | 3BC330           |